

1) To check whether a given number is even or not

```
num1 = int(input("Enter the number u want to check: "));  
if(num1 % 2 == 0):  
    print("It is an even number");  
else:  
    print("It is an odd number");
```

```
Python 3.12.3 (main, Mar 23 2026, 19:04:32) [GCC 13.3.0]  
Type "copyright", "credits" or "license" for more information.  
  
IPython 8.20.0 -- An enhanced Interactive Python.  
  
In [1]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/  
matlab/.config/spyder-py3')  
File <unknown>:3  
    if(num1 % 2 = 0):  
        ^  
SyntaxError: cannot assign to expression here. Maybe you meant '=='  
instead of '='?  
  
In [2]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/  
matlab/.config/spyder-py3')  
Enter the number u want to check: 20  
Traceback (most recent call last):  
  
  File /usr/lib/python3/dist-packages/spyder_kernels/py3compat.py:356 in  
compat_exec  
    exec(code, globals, locals)  
  
  File ~/.config/spyder-py3/temp.py:2  
    for i in num1:  
TypeError: 'int' object is not iterable  
  
In [3]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/  
matlab/.config/spyder-py3')  
Enter the number u want to check: 20  
It is an even number  
  
In [4]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/  
matlab/.config/spyder-py3')  
Enter the number u want to check: 13  
It is an odd number
```

2) To check whether a number is divisible or not

```
def is_prime(n):
    if n <= 1:
        return False;
    for i in range(2 , int(n**0.5) + 1):
        if n%i == 0:
            return False;
    return True;

num = int(input("Enter a number: "));

if is_prime(num):
    print(num,"is a prime number");
else:
    print(num,"It is not prime");
```

```
In [6]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/
matlab/.config/spyder-py3')
File <unknown>:5
    if n%i == 0;;
                ^
SyntaxError: invalid syntax

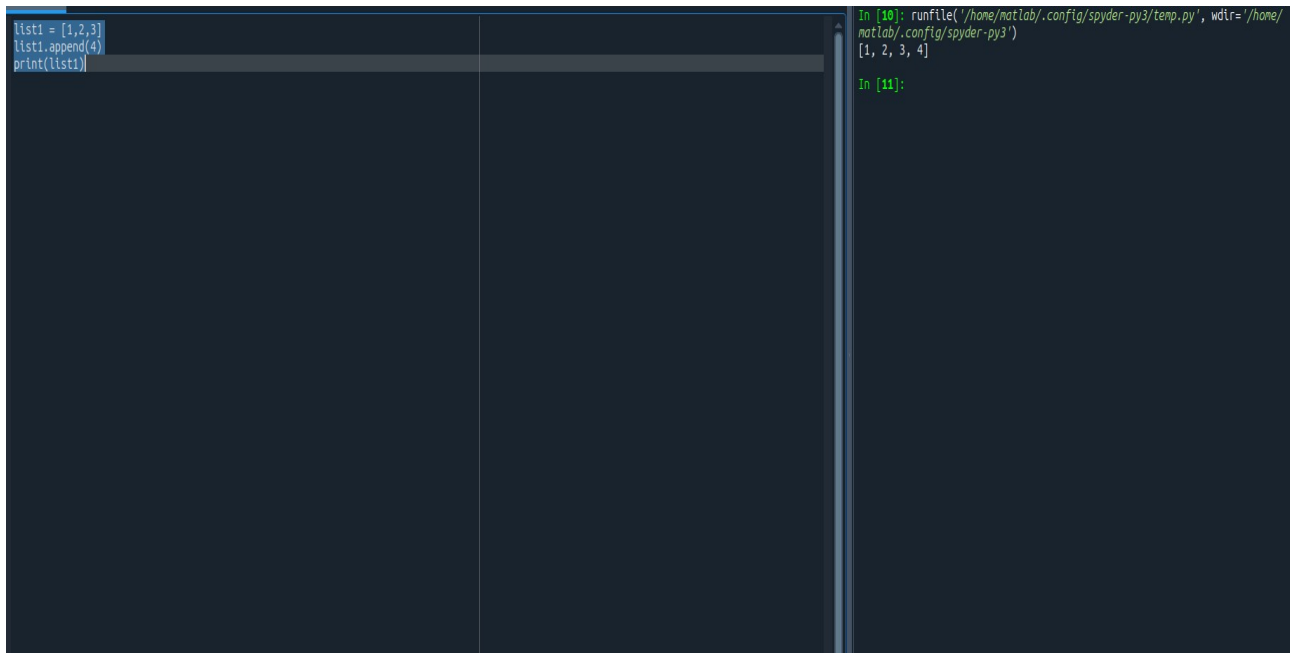
In [7]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/
matlab/.config/spyder-py3')
Enter a number: 3
3 is a prime number

In [8]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/
matlab/.config/spyder-py3')
Enter a number: 6
6 It is not prime

In [9]: |
```

3) Appending elements in a list

```
list1 = [1,2,3]  
list1.append(4)  
print(list1)
```



The screenshot shows the Spyder Python IDE interface. On the left, the editor contains the following code:

```
list1 = [1,2,3]  
list1.append(4)  
print(list1)
```

On the right, the IPython console shows the execution of the code:

```
In [10]: runfile('/home/natlab/.config/spyder-py3/temps.py', wdir='/home/  
natlab/.config/spyder-py3')  
[1, 2, 3, 4]  
  
In [11]:
```

4) Application of dictionary

```
students = {  
    "Name": "Amit",  
    "Age" : 20,  
    "Branch": "CSE"  
}
```

```
print(students["Name"]);  
print(students["Age"];
```

```
students = {  
    "Name": "Amit",  
    "Age" : 20,  
    "Branch": "CSE"  
}  
  
print(students["Name"]);  
print(students["Age"]);
```

```
In [10]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/  
matlab/.config/spyder-py3')  
[1, 2, 3, 4]
```

```
In [11]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/  
matlab/.config/spyder-py3')  
Traceback (most recent call last):
```

```
File /usr/lib/python3/dist-packages/spyder_kernels/py3compat.py:356 in  
compat_exec  
exec(code, globals, locals)
```

```
File ~/config/spyder-py3/temp.py:7  
print(students["name"]);
```

```
KeyError: 'name'
```

```
In [12]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/  
matlab/.config/spyder-py3')  
Amit
```

```
Traceback (most recent call last):
```

```
File /usr/lib/python3/dist-packages/spyder_kernels/py3compat.py:356 in  
compat_exec  
exec(code, globals, locals)
```

```
File ~/config/spyder-py3/temp.py:8  
print(student["Age"]);
```

```
NameError: name 'student' is not defined
```

```
In [13]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/  
matlab/.config/spyder-py3')  
Amit
```

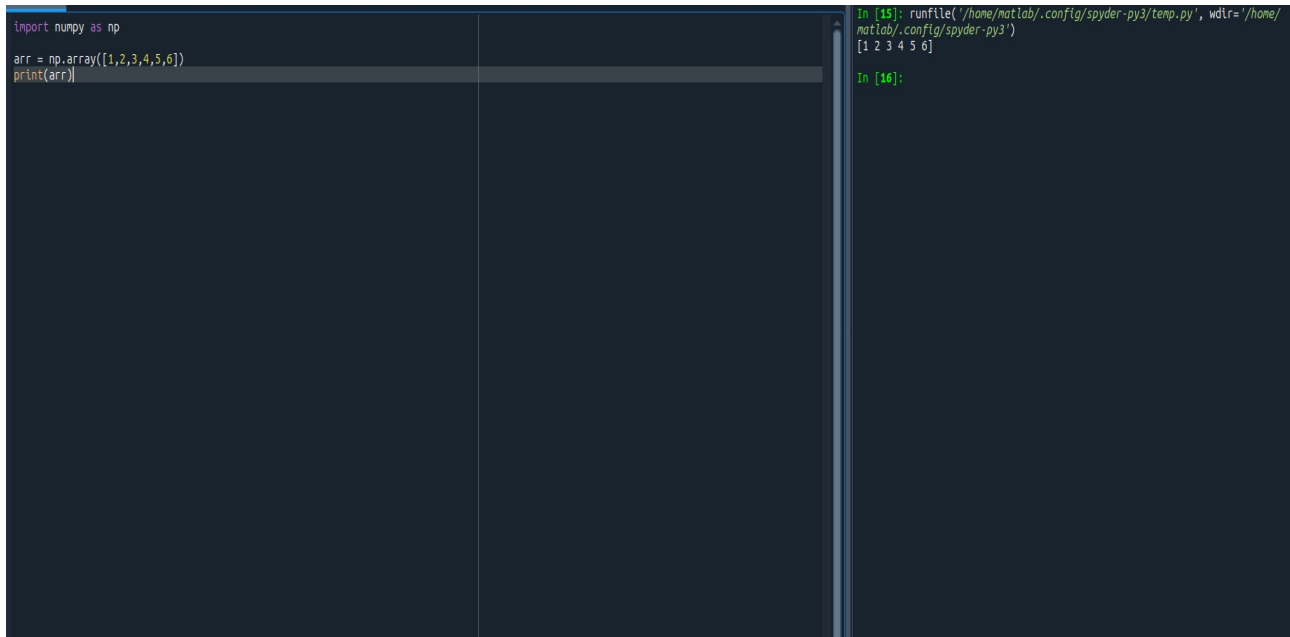
```
20
```

```
In [14]:
```

5) Numpy basics

```
import numpy as np
```

```
arr = np.array([1,2,3,4,5,6])  
print(arr)
```



The screenshot shows the Spyder IDE interface. The left pane contains the following code:

```
import numpy as np  
arr = np.array([1,2,3,4,5,6])  
print(arr)
```

The right pane shows the IPython console output for two input prompts:

```
In [15]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/  
matlab/.config/spyder-py3')  
[1 2 3 4 5 6]  
  
In [16]:
```

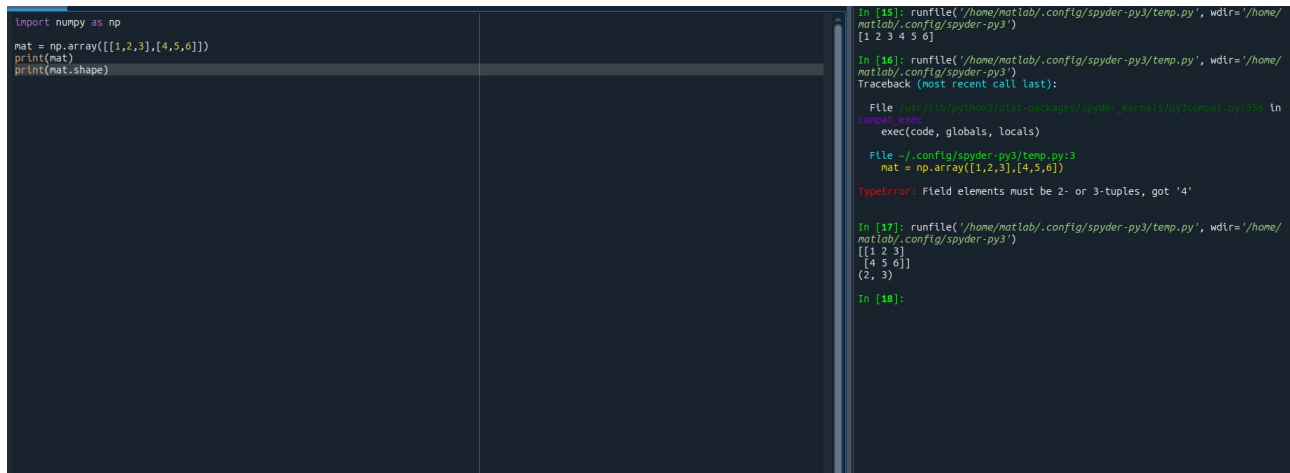
6) Matrix using numpy

import numpy as np

mat = np.array([[1,2,3],[4,5,6]])

print(mat)

print(mat.shape)



The screenshot shows the Spyder IDE interface. The left pane contains the following Python code:

```
import numpy as np
mat = np.array([[1,2,3],[4,5,6]])
print(mat)
print(mat.shape)
```

The right pane shows the IPython console output:

```
In [15]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/matlab/.config/spyder-py3')
[1 2 3 4 5 6]

In [16]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/matlab/.config/spyder-py3')
Traceback (most recent call last):
  File ~/lib/python3/dist-packages/spyder_kernels/py3compat.py:356: in
    exec(code, globals, locals)
  File ~/.config/spyder-py3/temp.py:3
    mat = np.array([1,2,3],[4,5,6])
TypeError: Field elements must be 2- or 3-tuples, got '4'

In [17]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/matlab/.config/spyder-py3')
[[1 2 3]
 [4 5 6]]
(2, 3)

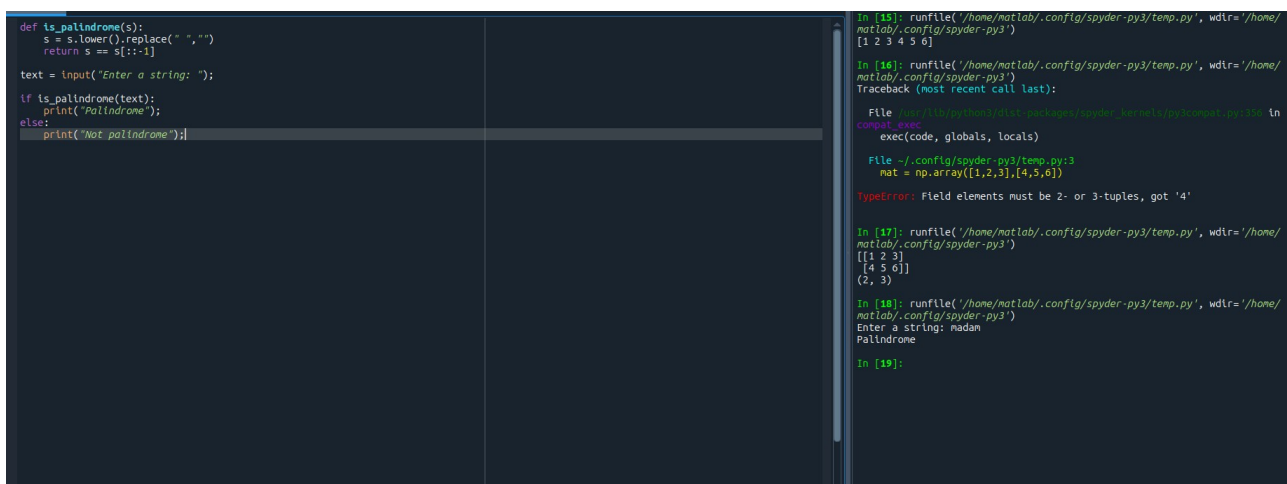
In [18]:
```

7) To find whether a word is a palindrome

```
def is_palindrome(s):
    s = s.lower().replace(" ", "")
    return s == s[::-1]
```

text = input("Enter a string: ");

```
if is_palindrome(text):
    print("Palindrome");
else:
    print("Not palindrome");
```



The screenshot shows the Spyder IDE interface. The left pane contains the following Python code:

```
def is_palindrome(s):
    s = s.lower().replace(" ", "")
    return s == s[::-1]

text = input("Enter a string: ");

if is_palindrome(text):
    print("Palindrome");
else:
    print("Not palindrome");
```

The right pane shows the IPython console output:

```
In [15]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/matlab/.config/spyder-py3')
[1 2 3 4 5 6]

In [16]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/matlab/.config/spyder-py3')
Traceback (most recent call last):
  File ~/lib/python3/dist-packages/spyder_kernels/py3compat.py:356: in
    exec(code, globals, locals)
  File ~/.config/spyder-py3/temp.py:3
    mat = np.array([1,2,3],[4,5,6])
TypeError: Field elements must be 2- or 3-tuples, got '4'

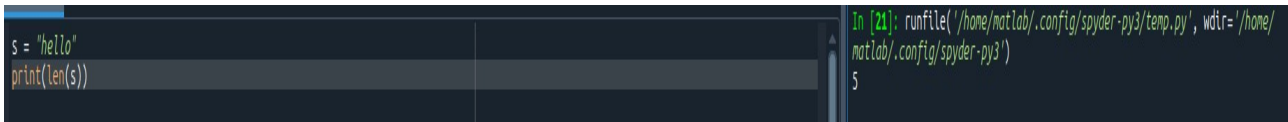
In [17]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/matlab/.config/spyder-py3')
[[1 2 3]
 [4 5 6]]
(2, 3)

In [18]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/matlab/.config/spyder-py3')
Enter a string: nadan
Palindrome

In [19]:
```

8) String implementation – length of a string

```
s = "hello"  
print(len(s))
```



```
s = "hello"  
print(len(s))
```

In [21]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/matlab/.config/spyder-py3')

5

9) string implementation – indexing

```
s = "hello"  
print(s[0:3:1]);
```



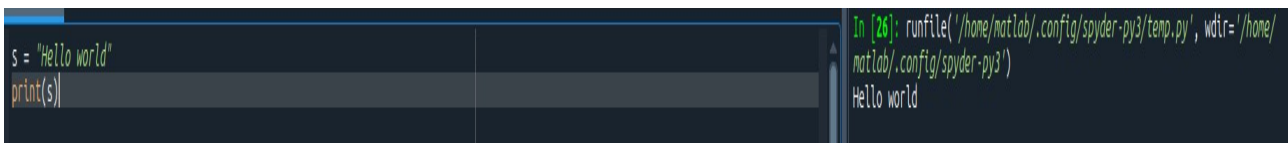
```
s = "hello"  
print(s[0:3:1]);
```

In [24]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/matlab/.config/spyder-py3')

hel

10) Basic of python – Hello world

```
s = "Hello world"  
print(s)
```



```
s = "Hello world"  
print(s)
```

In [26]: runfile('/home/matlab/.config/spyder-py3/temp.py', wdir='/home/matlab/.config/spyder-py3')

Hello world